Lecture 11 v2:
An Introduction to Binary Choice Models

25117 - Econometrics

Universitat Pompeu Fabra

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What we learned in the last lesson

- Panel data consist of observations on multiple (n) units – states, firms, people, and so forth – where each unit is observed at two or more time periods (T).
- Regression with unit fixed effects controls for unobserved variables that differ from one entity to the next but remain constant over time.
- Unit fixed effects regression can be estimated by including binary variables for $n-1$ entities plus the observable independent variables (the X's) and an intercept.
- Time fixed effects control for unobserved variables that are the same across entities but vary over time.
- A regression with time and entity fixed effects can be estimated by including binary variables for $n-1$ entities and binary variables for $T-1$ time periods plus the X's and an intercept.
- In panel data, variables are typically autocorrelated – that is, correlated over time within an entity. Clustered standard errors allow for this autocorrelation within entities.

Binary Choice Models

So far the dependent variable Y has been continuous:

- district-wide average test score
- traffic fatality rate

What if Y is binary?

- get into college, or not
- smoking, or not
- passing exams, or not
- mortgage application is accepted, or not

Example: Mortgage Denial and Race The Boston Fed HMDA Dataset

- Y – Is mortgage accepted?
- X – debt payments to income ratio (P/I ratio)
- other covariates – race of applicant, wealth, employment status, other loan, property characteristics

Binary Choice Models

To study the relationship between a set of variables x_i' and the dependent variable y_i, we need to extract information about:

$$E[y_i \mid x_i']$$

The conditional expectation in the familiar linear regression case is $E[y_i \mid x_i'] = x_i'\beta$.

In this topic, we will posit that the relationship between the vector of explanatory variables x_i' and the dependent variable is not necessarily linear. In particular, we define:

$$E[y_i \mid x_i'] = F(x_i'\beta)$$

So, a central element of this topic will be the choice of the distribution function F

Linear Probability Model

A natural starting point is the linear regression model with a single regressor:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

In the linear probability model (LPM), the predicted value of Y is interpreted as the predicted probability that $Y = 1$ and β_1 is the change in that predicted probability for a unit change in X.

When Y is binary,

$$\mathbb{E}(Y \mid X) = 1 \times Pr(Y = 1 \mid X) + 0 \times Pr(Y = 0 \mid X) = Pr(Y = 1 \mid X) = \beta_0 + \beta_1 X_i$$

So $\hat{Y}$ is the predicted probability that $Y = 1$ and β_1 is the difference in probability that $Y = 1$ when X increases by one unit.

$$\beta_1 = \frac{Pr(Y = 1 \mid X + \Delta X) - Pr(Y = 1 \mid X)}{\Delta X}$$

Linear Probability Model

Let's start by defining that $F(x_i'\beta) = x_i'\beta$. For example, if we have k explanatory variables, the equation we aim to estimate will be:

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + ... + \beta_k x_{k,i} + u_i$$

- **Parameter Interpretation**: β_j captures the change in $Pr(y_i = 1)$ when x_j increases by one unit.

$$\frac{\partial y_i}{\partial x_{i,j}} = \beta_j$$

- **Estimation:** The parameter vector $\beta = \{\alpha, \beta_1, ..., \beta_k\}$ is estimated using ordinary least squares (OLS).

$$\beta^{OLS} = \underset{\beta}{arg\ max} \sum_{i=1}^{n} (y_i - x_i'\beta)^2$$

Properties of the Linear Probability Model

- **Conditional Expectation**:

$$E[y_i \mid x_i] = Pr[y_i = 1 \mid x_i] = x_i'\beta$$

- **Total Probability**:

$$Pr[y_i = 0 \mid x_i] = 1 - x_i'\beta$$

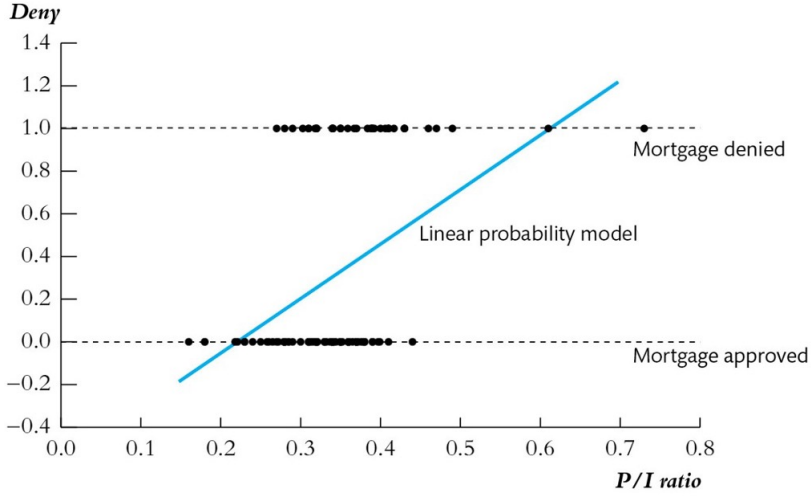
- **Expectation of Errors:**

$$Assumption: \quad E[u_i \mid x_i'] = 0$$

Linear Probability Model

- The linear probability model models $Pr(Y = 1 \mid X)$ as a linear function of X
- Is it a correct functional form?

→ Here, mortgage denial v. ratio of debt payments to income (P/I ratio) in a subset of the HMDA data set (n = 127)



Linear Probability Model

There advantages and disadvantages in modelling $Pr(Y = 1 \mid X)$ as a linear function of X

Pros

- Very simple to estimate and to interpret

Cons

- A LPM says that the change in the predicted probability for a given change in X is the same (percentage-point) for all values of X, but that does not make sense.
- LPM predicted probabilities can be larger than 1 and smaller than 0!

These disadvantages can be solved by using a nonlinear probability model...

$\longrightarrow$ probit and logit regressions

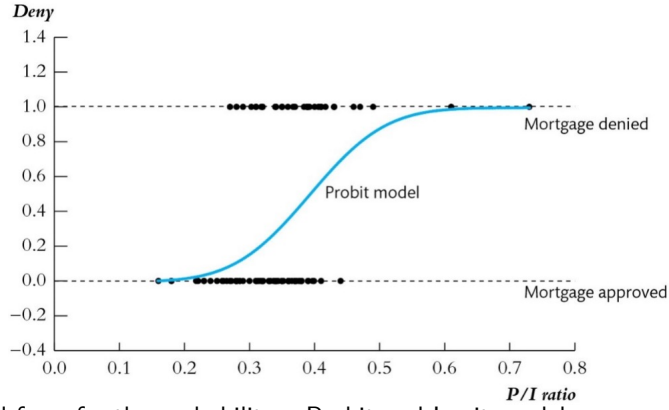
Nonlinear Models

The problem with the linear probability model is that it models the probability of $Y = 1$ as a linear function of X:

$$Pr(Y = 1 \mid X) = \beta_0 + \beta_1 X_i$$

Instead, we want:

- $Pr(Y = 1 \mid X)$ increases with X if β_1 is positive
- $0 \leq Pr(Y = 1 \mid X) \leq 1 \forall X$



This requires using a nonlinear functional form for the probability... Probit and Logit models satisfies these conditions

Nonlinear Models

So far, we have treated $Pr(y_i = 1 \mid x_i')$ linearly. Now, assume that this probability is given by a distribution function $F(x_i'\beta)$.

To introduce the distribution of the variable y_i, consider the following latent variable model:

$$y_i^* = x_i'\beta + \varepsilon_i$$

If in this latent variable model y_i^* is positive, then $y_i = 1$, and if y_i^* is negative, then $y_i = 0$. In this case:

$$Pr(y_i = 1 \mid x_i) = Pr(y_i^* > 0 \mid x_i) = Pr(\varepsilon_i > -x_i'\beta \mid x_i) = F(x_i'\beta)$$

Therefore, when choosing the distribution function of errors in the latent model, we are determining how the probability of choosing an alternative behaves.

Probit Model

Probit regression models the probability that $Y = 1$ using the cumulative standard *normal* distribution function, $\Phi(z)$, evaluated at $z = \beta_0 + \beta_1 X_1$. So the probit regression model reads:

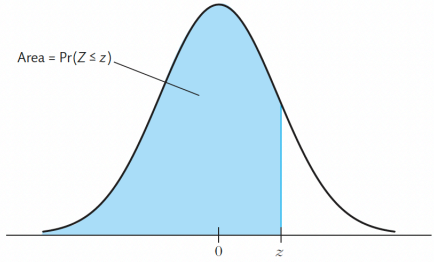
$$Pr(Y = 1 \mid X_i) = \Phi(\beta_0 + \beta_1 X_i) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\beta_0 + \beta_1 X_i} e^{-t^2/2} dt$$

Example: Suppose that $\beta_0 = -3.5$ and $\beta_1 = 2$. Then

$$Pr(Y = 1 \mid X = .61) = \Phi(-3.5 + 2 \times .61) = \Phi(-2.28) = 0.0113$$

$Pr(Y = 1 \mid X = .61)$ corresponds to the area under the standard normal density to left of $z = -2.28$

Probit Model



Second Decimal Value of *z*

z	0	1	2	3	4	5	6	7	8	9
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183

Logit Model

Logit regression models the probability that $Y = 1$ using the cumulative standard *logistic* distribution function, $F(z)$, evaluated at $z = \beta_0 + \beta_1 X_1$. So the logit regression model reads:

$$Pr(Y = 1 \mid X_i) = F(\beta_0 + \beta_1 X_i) = \frac{e^{\beta_0 + \beta_1 X_i}}{1 + e^{\beta_0 + \beta_1 X_i}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_i)}}$$

Example: Suppose (again) that $\beta_0 = -3.5$ and $\beta_1 = 2$. Then

$$Pr(Y = 1 \mid X = .61) = F(-3.5 + 2 \times .61) = \frac{1}{1 + e^{2.28}} = 0.0928$$

$Pr(Y = 1 \mid X = .61)$ corresponds to the area under the standard *logistic* density to left of $z = -2.28$

Because logit and probit use different probability functions, the coefficients βs are different and have different interpretations...

Marginal Effects

The *probit* and *logit* estimates provide the vector β, but not the marginal effects, which are:

$$\frac{\partial \Phi(x_i'\beta)}{\partial x_k} = \beta_k \phi(x_i'\beta) \quad \textit{Probit}$$

$$\frac{\partial \frac{e^{x_i'\beta}}{1+e^{x_i'\beta}}}{\partial x_k} = \beta_k Pr(y_i = 1)(1 - Pr(y_i = 1)) \quad \textit{Logit}$$

Therefore, to evaluate the marginal effects, we need to provide values to x_i'. In Stata, we calculate the marginal effects using the `margins` function after estimating the corresponding model. With this function, Stata calculates the average marginal effects, i.e.:

$$\frac{1}{n} \sum_i \frac{\partial E[y_i \mid x_i]}{\partial x_k}$$

Marginal Effects

A distinction must be made between:

1. Average marginal effects: obtain the marginal effect on each individual and then calculate the average of the marginal effects
$$\frac{1}{n}\sum_i \frac{\partial E[y_i \mid x_i]}{\partial x_k}$$
2. Marginal effects calculated at the mean: obtain the marginal effect by evaluating the expression at the mean of the explanatory variables:
$$\frac{\partial E[y_i \mid x_i]}{\partial x_k} \Big|_{x_i=\bar{x}}$$

   Since $E[y_i \mid x_i]$ is not linear in x_i, the two calculations yield different results.

Marginal Effects (Logit)

```
. margins, dydx(*)

Average marginal effects                        Number of obs     =        753
Model VCE    : OIM

Expression   : Pr(inlf), predict()
dy/dx w.r.t. : nwifeinc age educ exper kidslt6 kidsge6
```

	dy/dx	Delta-method Std. Err.	z	P>|z|	[95% Conf.	Interval]
nwifeinc	-.0038118	.0014824	-2.57	0.010	-.0067172	-.0009064
age	-.0157194	.0023808	-6.60	0.000	-.0203856	-.0110532
educ	.0394965	.0072947	5.41	0.000	.0251992	.0537939
exper	.0254254	.0022364	11.37	0.000	.0210421	.0298088
kidslt6	-.2577537	.0319416	-8.07	0.000	-.3203581	-.1951492
kidsge6	.0107348	.013333	0.81	0.421	-.0153974	.0368671

Marginal Effects (Logit)

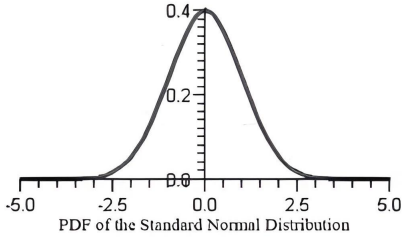
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. margins, dydx(*) atmeans

Conditional marginal effects                    Number of obs     =        753
Model VCE    : OIM

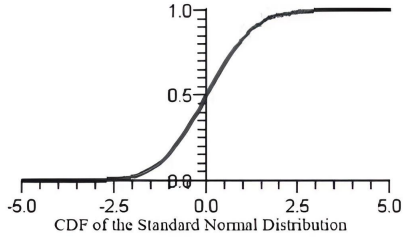
Expression   : Pr(inlf), predict()
dy/dx w.r.t. : nwifeinc age educ exper kidslt6 kidsge6
at           : nwifeinc          =    20.12896 (mean)
               age               =    42.53785 (mean)
               educ              =    12.28685 (mean)
               exper             =    10.63081 (mean)
               kidslt6           =    .2377158 (mean)
               kidsge6           =    1.353254 (mean)
```

	Delta-method dy/dx	Std. Err.	z	P>|z|	[95% Conf.	Interval]
nwifeinc	-.0049664	.0019585	-2.54	0.011	-.0088051	-.0011277
age	-.0204807	.0034374	-5.96	0.000	-.0272179	-.0137435
educ	.0514599	.0101878	5.05	0.000	.0314921	.0714277
exper	.0322966	.0033005	9.79	0.000	.0258277	.0387655
kidslt6	-.3358267	.0480192	-6.99	0.000	-.4299426	-.2417108
kidsge6	.0139864	.0174041	0.80	0.422	-.0201251	.0480979

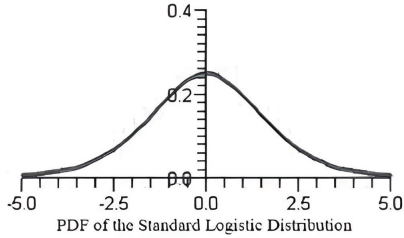
So... Probit or Logit?



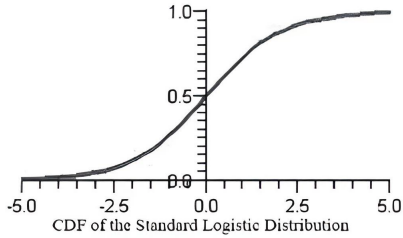
PDF of the Standard Normal Distribution



CDF of the Standard Normal Distribution

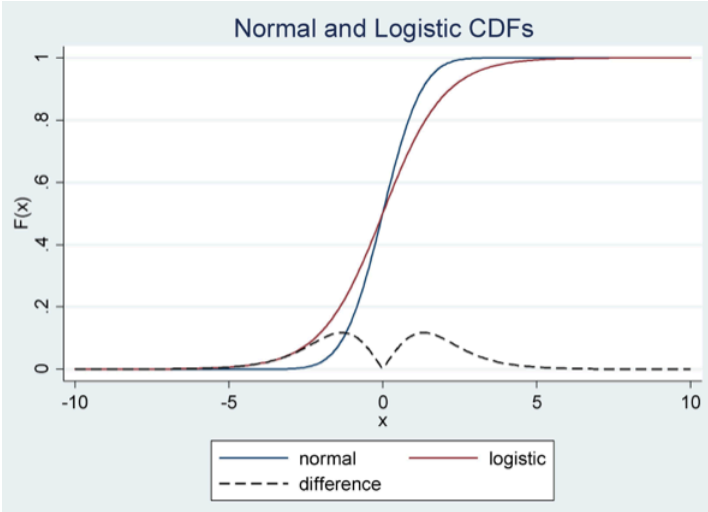


PDF of the Standard Logistic Distribution



CDF of the Standard Logistic Distribution

Solution to Fitted Probabilities



So... Probit or Logit?

Why do we have two different models? The main reason is historical: logit is computationally faster and easier, but that doesn't matter nowadays with modern computation softwares.

Now, the logistic distribution displays fatter tails (i.e the probit curve approaches the axes more quickly than the logit curve.)

This matters for extremes, but in practice, logit and probit are very similar when the binary outcome is decently balanced (perfectly balanced 50% 0s, 50% 1s)

This is the case in the vast majority of the applications...

Logit models tend to be slightly more used in practice because the βs are easier to interpret: they are in log-odds units ($ln(\frac{Pr(Y=1|X)}{1-Pr(Y=1|X)})$).

Log-Odds-Ratio

Suppose that:

$$\frac{Pr(y_i = 1)}{Pr(y_i = 0)} = 1.1 \implies \textit{The probability of choosing 1 is 10\% greater than choosing 0}$$

In terms of log-odds ratio (assuming odds ratio is close to 1):

$$ln(Pr(y_i = 1)) - ln(Pr(y_i = 0)) \approx 0.1$$

Also indicates that the probability of choosing 1 over 0 is 10% greater.

- Important: if $\beta_k > 0$:

$$\frac{\partial odds}{\partial x_k} > 0 \quad \& \quad \frac{\partial log - odds}{\partial x_k} > 0$$

Why log-odds units?

$$
\begin{aligned}
Pr(Y = 1 \mid X_i) &= F(\beta_0 + \beta_1 X_i) = \frac{e^{\beta_0 + \beta_1 X_i}}{1 + e^{\beta_0 + \beta_1 X_i}} \\
\Leftrightarrow \quad & \frac{1}{Pr(Y = 1 \mid X_i)} = \frac{1 + e^{\beta_0 + \beta_1 X_i}}{e^{\beta_0 + \beta_1 X_i}} = 1 + \frac{1}{e^{\beta_0 + \beta_1 X_i}} \\
\Leftrightarrow \quad & \frac{1 - Pr(Y = 1 \mid X_i)}{Pr(Y = 1 \mid X_i)} = \frac{1}{e^{\beta_0 + \beta_1 X_i}} \\
\Leftrightarrow \quad & \frac{Pr(Y = 1 \mid X_i)}{1 - Pr(Y = 1 \mid X_i)} = e^{\beta_0 + \beta_1 X_i} \\
\Leftrightarrow \quad & ln\left[\frac{Pr(Y = 1 \mid X_i)}{1 - Pr(Y = 1 \mid X_i)}\right] = \beta_0 + \beta_1 X_i
\end{aligned}
$$

Using the preceding example, a 10%-point increase in the P/I ratio, *ceteris paribus*, would yield a 0.537 ($.1 \times 5.37$) increase in the log odds of being denied a mortgage... Put differently, the odds (p/(1-p)) of being denied a mortgage would increase by 71% ($e^{0.537} = 1.71$) !

Maximum Likelihood Estimation

Estimation and Inference in the Logit and Probit Models

In your previous lectures, you were assuming regression models that are linear or nonlinear (e.g., polynomials) functions of the independent variables but are linear functions of the unknown coefficients (parameters) $\rightarrow$ can be estimated by OLS.

In contrast, the probit and logit regression functions are nonlinear functions of the coefficients $\rightarrow$ cannot be estimated by OLS $\rightarrow$ NEW METHOD: **Maximum Likelihood**

Maximum Likelihood Estimation (MLE) in econometrics is a method for estimating the parameters of a statistical model by maximizing the likelihood function.

The likelihood function represents the probability of observing the given sample data under different parameter values.

Formally, MLE seeks to find the parameter values that maximize the likelihood function, or equivalently, minimize the negative log-likelihood.

Estimation and Inference in the Logit and Probit Models

This process is akin to searching for the parameter values that make the observed data most probable, treating the sample as a realization of a probability distribution determined by the model.

The theory of ML says that the ML estimator $\hat{p}^{MLE}$ is the most efficient estimator of p — of *all* consistent estimators! — at least for large n. (Much stronger than the Gauss-Markov theorem!).

For this reason, the MLE is the primary estimator used for models in which the parameters (coefficients) enter nonlinearly.

In large samples, the MLE is:

- Consistent
- Normally distributed
- Efficient (has the smallest variance of all consistent estimators)

Bernoulli example

- Suppose that we know that the following ten numbers were simulated using a Bernoulli distribution: $0, 0, 0, 1, 1, 0, 0, 1, 0, 1$.
- Denote them by $y_1, y_2, \ldots, y_{10}$. So $y_1 = 0$ and $y_{10} = 1$
- The probability of 1 is p, while the probability of 0 is $(1 - p)$.
- Recall that the pdf of a Bernoulli random variable is $f(y; p) = p^y(1 - p)^{1-y}$, where $y \in \{0, 1\}$
- We want to find the p that was used to simulate the ten numbers.
- **All we know** is that 1) they are drawn from a Bernoulli distribution and 2) independently from each other

Bernoulli example (contd.)

- Since we know the pdf that generated the numbers is Bernoulli, we know that the probability of the *first* number is $f(y_1; p) = p^{y_1}(1-p)^{1-y_1}$
- ... and we know that the probability of the *second* number is $f(y_2; p) = p^{y_2}(1-p)^{1-y_2}$, etc.
- We could replace the y_i with the actual numbers. For example, the first one is $y_1 = 0$, so the probability is just $(1-p)$.
- What we do not know is **the value of the parameter** p
- Since we know that they are independent, we could also write down the probability of observing all ten numbers. That is, their **joint probability**.
- Since they are independent, their joint probability distribution is the multiplication of the ten pdfs: $P(A \cap B) = P(A) \times P(B)$

Bernoulli example (contd.)

- Since they are independent, we could write down the joint probability or the **likelihood** (L) of seeing those 10 numbers as:

$$L(p) = \prod_{i=1}^{10} p^{y_i}(1-p)^{1-y_i}$$

- We want to find the $\hat{p}$ that maximizes the likelihood function $L(p)$, i.e., the probability of observing all ten y_i

- Taking that derivative is complicated because we would need to use the chain rule several times so we take the log to simplify the notation. The log function is a monotonic transformation, it won't change the optimal $\hat{p}$:

$$\ln L(p) = \sum_{i=1}^{n} y_i \ln(p) + \sum_{i=1}^{n} (1-y_i) \ln(1-p)$$

Bernoulli example (contd.)

- In this case, the log-likelihood function simplifies to

$$\ln L(p) = n\bar{y}\ln(p) + n(1-\bar{y})\ln(1-p)$$

- Taking the derivative with respect to p and setting it to zero, we find the MLE estimator of p:

$$\frac{\partial \ln(p)}{\partial p} = \frac{\bar{y}}{p} - \frac{1-\bar{y}}{1-p} = 0$$

- Solving, we find that $\hat{p} = \bar{y} = \sum_{i=1}^{n} y_i/n$.

- So that's the MLE estimator of p. This is saying more or less the obvious: our best guess for the p that generated the data is the proportion of 1s; in this case $p = 0.4$

Maximum Likelihood Estimation (MLE)

Analogously, the probit likelihood function is the joint density of $Y_1, Y_2, \ldots, Y_n$ given $X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n$ treated as function of the unknown parameters β_0, β_1 (for only one regressor):

$$\begin{aligned} f(\beta_0, \beta_1; Y_1, Y_2, \ldots, Y_n \mid X_1, X_2, \ldots, X_n) &= \{[\Phi(\beta_0 + \beta_1 x_1)]^{y_1} \times [1 - \Phi(\beta_0 + \beta_1 x_1)]^{1-y_1}\} \\ &\times \{[\Phi(\beta_0 + \beta_1 x_2)]^{y_2} \times [1 - \Phi(\beta_0 + \beta_1 x_2)]^{1-y_2}\} \\ &\times \ldots \\ &\times \{[\Phi(\beta_0 + \beta_1 x_n)]^{y_n} \times [1 - \Phi(\beta_0 + \beta_1 x_n)]^{1-y_n}\} \end{aligned}$$

The parameters $\hat{\beta}_0^{MLE}, \hat{\beta}_1^{MLE}$ maximize this likelihood function!

We can easily extend this to multiple regressors and/or the logistic regression model...

Maximum Likelihood Estimation (MLE)

The logit likelihood function is the joint density of $Y_1, Y_2, \ldots, Y_n$ given $X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n$ treated as function of the unknown parameters β_0, β_1 (for only one regressor):

$$\begin{aligned} f(\beta_0, \beta_1; Y_1, Y_2, \ldots, Y_n \mid X_1, X_2, \ldots, X_n) &= \{[F(\beta_0 + \beta_1 x_1)]^{y_1} \times [1 - F(\beta_0 + \beta_1 x_1)]^{1-y_1}\} \\ &\times \{[F(\beta_0 + \beta_1 x_2)]^{y_2} \times [1 - F(\beta_0 + \beta_1 x_2)]^{1-y_2}\} \\ &\times \ldots \\ &\times \{[F(\beta_0 + \beta_1 x_n)]^{y_n} \times [1 - F(\beta_0 + \beta_1 x_n)]^{1-y_n}\} \end{aligned}$$

In both cases, we cannot solve for the maximum explicitly! So, the MLE must be maximized using numerical methods (dynamic optimization algorithms).

Model Selection Criteria

When comparing two models that contain the same set of variables, we only need to examine the values of the likelihood functions. The higher, the better the model fits.

There are different methods for choosing between models that do not have the same explanatory variables:

1. Likelihood Ratio Test
2. Number of correct predictions
3. McFadden's R^2
4. Akaike Information Criterion (AIC)
5. Weighted Residual Sum of Squares

Model Selection Criteria

- Likelihood Ratio Test:

$$LR = 2[\mathcal{L}(\hat{\beta}_N) - \mathcal{L}(\hat{\beta}_R)] \sim \chi^2_q$$

  Where N and R represent the unrestricted and restricted (i.e., without explanatory variables) models respectively, and q is the difference in parameters. If we reject H_0, the unrestricted model will be better.

- Number of correct predictions: we assign $\hat{y}_i = 1$ if $\hat{P}_i > a$ and count the number of incorrect predictions from the model $\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$

- $WSSR = \sum_{i=1}^{n} \frac{(y_i - \hat{P}_i)^2}{\hat{P}_i(1-\hat{P}_i)}$. The model that minimizes this is preferable.

Model Selection Criteria

- Akaike Information Criterion (AIC): $AIC = -ln(\mathcal{L}) + K$, where K is the number of estimated parameters. The smaller the AIC, the better.

- McFadden's R^2: $R^2 = 1 - \frac{ln(\mathcal{L}_N)}{ln(\mathcal{L}_R)} \in [0, 1]$, where L_R is the value of the likelihood function when only a constant is included, and L_N is that of the unrestricted model.

Application: Women's Participation in the Labor Market

Dependent Variable $y_i \rightarrow inlf \in \{0, 1\}$ (0 if the woman is not in the labor market, 1 if she is)

Explanatory Variables x_i' : ($nwifeinc$, $educ$, exp, exp^2, age, $kidslt6$, $kidsge6$)

- LPM:

$$E[infl_i \mid x_i] = x_i'\beta$$

- Probit:

$$E[infl_i \mid x_i] = \Phi(x_i'\beta)$$

- Logit:

$$E[infl_i \mid x_i] = \frac{e^{x_i'\beta}}{1 + e^{x_i'\beta}}$$

LPM

```
. reg inlf nwifeinc age educ c.exper##c.exper kidslt6 kidsge6, r

Linear regression                               Number of obs     =        753
                                                F(7, 745)         =      62.48
                                                Prob > F          =     0.0000
                                                R-squared         =     0.2642
                                                Root MSE          =     .42713

------------------------------------------------------------------------------
             |               Robust
        inlf |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
    nwifeinc |  -.0034052   .0015249    -2.23   0.026    -.0063988   -.0004115
         age |  -.0160908    .002399    -6.71   0.000    -.0208004   -.0113812
        educ |   .0379953    .007266     5.23   0.000     .023731    .0522596
       exper |   .0394924     .00581     6.80   0.000     .0280864   .0508983
             |
c.exper#c.exper |  -.0005963   .00019    -3.14   0.002    -.0009693   -.0002233
             |
     kidslt6 |  -.2618105   .0317832    -8.24   0.000    -.3242058   -.1994152
     kidsge6 |   .0130122   .0135329     0.96   0.337    -.013555    .0395795
       _cons |   .5855192   .1522599     3.85   0.000     .2866098   .8844287
------------------------------------------------------------------------------
```

Probit

```
. probit inlf nwifeinc age educ c.exper##c.exper kidslt6 kidsge6

Iteration 0:   log likelihood =  -514.8732
Iteration 1:   log likelihood = -402.06651
Iteration 2:   log likelihood = -401.30273
Iteration 3:   log likelihood = -401.30219
Iteration 4:   log likelihood = -401.30219

Probit regression                                 Number of obs     =        753
                                                  LR chi2(7)        =     227.14
                                                  Prob > chi2       =     0.0000
Log likelihood = -401.30219                       Pseudo R2         =     0.2206

------------------------------------------------------------------------------
        inlf |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
    nwifeinc |  -.0120237   .0048398    -2.48   0.013    -.0215096   -.0025378
         age |  -.0528527   .0084772    -6.23   0.000    -.0694678   -.0362376
        educ |   .1309047   .0252542     5.18   0.000     .0814074    .180402
       exper |   .1233476   .0187164     6.59   0.000     .0866641    .1600311
             |
c.exper#c.exper |  -.0018871      .0006    -3.15   0.002    -.003063   -.0007111
             |
     kidslt6 |  -.8683285   .1185223    -7.33   0.000    -1.100628   -.636029
     kidsge6 |    .036005   .0434768     0.83   0.408    -.049208    .1212179
       _cons |   .2700768    .508593     0.53   0.595    -.7267473    1.266901
------------------------------------------------------------------------------
```

Logit

```
. logit inlf nwifeinc age educ c.exper##c.exper kidslt6 kidsge6

Iteration 0:   log likelihood =  -514.8732
Iteration 1:   log likelihood = -402.38502
Iteration 2:   log likelihood = -401.76569
Iteration 3:   log likelihood = -401.76515
Iteration 4:   log likelihood = -401.76515

Logistic regression                             Number of obs     =        753
                                                LR chi2(7)        =     226.22
                                                Prob > chi2       =     0.0000
Log likelihood = -401.76515                     Pseudo R2         =     0.2197
```

inlf	Coef.	Std. Err.	z	P>|z|	[95% Conf.	Interval]
nwifeinc	-.0213452	.0084214	-2.53	0.011	-.0378509	-.0048394
age	-.0880244	.014573	-6.04	0.000	-.116587	-.0594618
educ	.2211704	.0434396	5.09	0.000	.1360303	.3063105
exper	.2058695	.0320569	6.42	0.000	.1430391	.2686999
c.exper#c.exper	-.0031541	.0010161	-3.10	0.002	-.0051456	-.0011626
kidslt6	-1.443354	.2035849	-7.09	0.000	-1.842373	-1.044335
kidsge6	.0601122	.0747897	0.80	0.422	-.086473	.2066974
_cons	.4254524	.8603697	0.49	0.621	-1.260841	2.111746

Summary and Conclusion

- If Y_i is binary, $\mathbb{E}(Y \mid X) = Pr(Y = 1 \mid X)$
- Three models:
  - linear probability model (linear multiple regression)
  - probit (cumulative standard normal distribution)
  - logit (cumulative standard logistic distribution)
- LPM, probit, logit all produce predicted probabilities
- Effect of ΔX is the change in conditional probability that $Y = 1$. But for logit and probit, this depends on the initial X
- Probit and logit are estimated via maximum likelihood
  - Coefficients are normally distributed for large n
  - Large-n hypothesis testing, conf. intervals is as usual

  There are many models for limited dependent variables, other than binary variables, in econometric applications.
  - Discrete choice or multiple choice data (multinomial probit/logit)
  - Censored Data Models (Tobin's Tobit)
  - Truncated Data Models (Heckman's Sample Selection)
  - Count Data Models (Poisson and Negative Binomial)
  - Ordered response data (Ordered probit/logit)

Material I

- **Textbooks:**
  - Introduction to Econometrics, 4th Edition, Global Edition, by Stock and Watson – Chapters 11